

1. CR-010: Student Profile Image Upload for the Admin Student Management

Change Request Details

Field	Details
Change Request ID	CR-010
Title	Student Profile Image Upload for the Admin Student Management
Priority	Medium
Difficulty	Medium
Change Type	Frontend + Backend Enhancement
Affected Version	Student Management System v1.0 baseline
Dependency	Existing Student Management and Add Student functionality

2. Business Requirement

Currently, administrators can add and manage student information such as registration number, name, email address, phone number, and other existing student details. However, the system does not provide an option to upload or display a student's profile image.

When managing a large number of students, a profile image can help administrators identify students more easily and improve the usability of the student management system.

Therefore, the system should allow administrators to upload a profile image when adding a new student. The system should support commonly used image formats such as **PNG, JPG, and JPEG**.

The uploaded image should be associated with the correct student and displayed in the student list or student details section.

3. Current Behaviour

- The administrator can access the student management section.
- The administrator can add a new student using the existing Add Student form.
- Existing student information such as name, registration number, email, and phone number can be entered.
- The Add Student form does not currently contain a profile image upload field.
- Students are displayed without profile images.
- No image file validation is currently available.
- Student records are stored without a profile image path.
- Existing student CRUD operations use the existing `student_id`.

4. Requested Change

Add a **Student Profile Image Upload** facility to the administrator's Add Student form.

The implementation must:

- Provide a profile image upload field in the Add Student form.
- Allow administrators to select a student image from their computer.
- Support **PNG, JPG, and JPEG** image formats.
- Reject unsupported file formats.
- Validate the selected file before uploading.
- Display a suitable validation message when an unsupported file is selected.
- Store the uploaded image securely on the server.
- Store the image path/reference with the corresponding student record.
- Display the uploaded student image in the student list or student details page.
- Preserve the existing Add Student functionality.
- Preserve existing Edit and Delete functionality.
- Ensure the uploaded image is associated with the correct `student_id`.
- Maintain existing administrator authentication and authorization.
- Prevent unsafe file types from being uploaded.
- Provide a suitable default/placeholder image when no profile image is available.

5. Scope

In Scope

- Add Profile Image field to the Admin Add Student form.
- PNG image upload.
- JPG image upload.
- JPEG image upload.
- Client-side file type validation.
- Server-side file type validation.
- Image upload handling in the backend.
- Saving the image path/reference.
- Displaying the student profile image.
- Handling students without an image.
- Error messages for invalid image files.
- Preservation of existing Student CRUD functionality.
- Preservation of existing administrator authentication and authorization.

Out of Scope

- Video upload.
- PDF upload.
- Document upload.
- GIF upload.
- Multiple images for one student.
- Image editing or advanced image cropping.

- Changes to administrator authentication.
- Changes to student authorization.
- Changes to unrelated student management features.
- Image search functionality.
- Changes to authentication or authorization.

6. Expected / Potentially Affected Files

The following files may be affected depending on the existing project structure:

- `frontend/src/pages/DashboardPage.jsx`
- Existing Add Student component/page
- Existing dashboard stylesheet/CSS file
- Existing backend student controller
- Existing backend student route
- Existing upload configuration/middleware
- `database/schema.sql`

Additional File

An upload directory may be introduced:

`backend/uploads/students/`

This directory is used to store uploaded student profile images.

Any additional files modified during implementation should be identified and justified in the Implementation Report.

7. Functional & Security Requirements

FR-001

The Add Student form shall provide a profile image upload field.

FR-002

The system shall allow administrators to upload PNG images.

FR-003

The system shall allow administrators to upload JPG images.

FR-004

The system shall allow administrators to upload JPEG images.

FR-005

The system shall reject unsupported file formats.

FR-006

The system shall display a clear validation message when an invalid file is selected.

FR-007

The uploaded image shall be associated with the correct student record.

FR-008

The uploaded profile image shall be displayed in the student list or student details section.

FR-009

The system shall provide a suitable placeholder when a student does not have a profile image.

FR-010

Existing Edit and Delete functionality shall continue to work correctly.

Security Requirements

SEC-001

Existing administrator authentication and authorization shall remain unchanged.

SEC-002

Only authorized administrators shall be allowed to upload student profile images.

SEC-003

The system shall accept only permitted image formats: PNG, JPG, and JPEG.

SEC-004

Unsafe file types such as .exe, .js, .php, .html, .pdf, and other unsupported files shall not be accepted as student profile images.

SEC-005

Uploaded files shall not be rendered using unsafe HTML.

SEC-006

The image upload functionality shall not expose additional sensitive student information.

SEC-007

The existing student `student_id` shall be used to associate the uploaded image with the correct student.

8. Acceptance Criteria

The CR is successfully implemented when:

AC-001

The Add Student form displays a Profile Image upload field.

AC-002

Administrators can select a PNG image.

AC-003

Administrators can select a JPG image.

AC-004

Administrators can select a JPEG image.

AC-005

Unsupported file formats are rejected.

AC-006

A clear error message is displayed when an invalid file is selected.

AC-007

A valid profile image is successfully uploaded when a student is created.

AC-008

The uploaded image is associated with the correct student.

AC-009

The uploaded image is displayed correctly in the student list or student details page.

AC-010

Students without an image display a default/placeholder image.

AC-011

Existing Edit and Delete operations continue to affect the correct student.

AC-012

Existing administrator authentication and authorization continue to work.

AC-013

No unsupported or unsafe file type can be uploaded through the profile image field.

9. Expected Outcome

After implementation, administrators will be able to add a student together with a profile image.

The Add Student process will become:

```
Admin Login
  ↓
Admin Dashboard
  ↓
Add Student
  ↓
Enter Student Details
  ↓
Select Profile Image
  ↓
Validate Image
  ↓
PNG / JPG / JPEG
  ↓
Upload Image
  ↓
Create Student Record
  ↓
Display Student Profile Image
```

Add Student

Registration Number: [ST001]

First Name: [Kamal]

Last Name: [Perera]

Email: [kamal@gmail.com]

Phone: [077xxxxxxx]

Profile Image:
[Choose File]

Supported formats:
PNG, JPG, JPEG

[Add Student]

After successful registration:

Student List

Image	Student ID	Name	Email
[Photo]	ST001	Kamal Perera	kamal@...
[Photo]	ST002	Nimal Silva	nimal@...

The feature will improve student identification and usability while preserving the existing student management, authentication, authorization, and CRUD functionality.

10. Implementation Report Requirement

After implementation, the Final Exam Software Change Implementation Report shall include:

- Implementation summary.
- Business reason for introducing student profile images.
- Affected files.
- Database changes, if applicable.
- Frontend implementation details.
- Backend implementation details.
- Image validation details.

- Supported image formats.
- Security considerations.
- Test cases and results.
- Acceptance criteria verification.
- Relevant screenshots/evidence.
- Final status.
- Outstanding issues, if any.

11. Suggested Test Cases

Test Case ID	Test Scenario	Test Data	Expected Result	Status
TC-001	Add student without image	Valid student details	Student added successfully	Pass
TC-002	Upload PNG	student.png	Image accepted	Pass
TC-003	Upload JPG	student.jpg	Image accepted	Pass
TC-004	Upload JPEG	student.jpeg	Image accepted	Pass
TC-005	Upload PDF	student.pdf	File rejected	Pass
TC-006	Upload EXE	student.exe	File rejected	Pass
TC-007	Upload unsupported file	document.txt	File rejected	Pass
TC-008	Add student with valid image	Student details + JPG	Student and image saved	Pass
TC-009	Display uploaded image	Existing student	Correct image displayed	Pass
TC-010	Student without image	Student details only	Placeholder displayed	Pass
TC-011	Edit student	Existing student	Correct student updated	Pass
TC-012	Delete student	Existing student	Correct student deleted	Pass
TC-013	Unauthorized upload attempt	Non-admin user	Access denied	Pass

Note: In the actual Implementation Report, change the `Status` column from `Pass` to the real result after performing each test.

12. Evidence / Screenshot Requirements

The following screenshots should be captured for the Implementation Report:

Evidence 1 – Add Student Before Change

Show the original Add Student form without the image field.

Evidence 2 – Add Student After Change

Show:

Profile Image
[Choose File]

Evidence 3 – PNG Upload

Select a .png image and show that it is accepted.

Evidence 4 – JPG Upload

Select a .jpg image and show that it is accepted.

Evidence 5 – JPEG Upload

Select a .jpeg image and show that it is accepted.

Evidence 6 – Invalid File Validation

Try uploading a PDF or another unsupported file and capture the validation message.

Example:

Only PNG, JPG and JPEG images are allowed.

Evidence 7 – Successful Student Creation

Show the success message after adding a student with an image.

Evidence 8 – Student List

Show the uploaded profile image beside the correct student.

Evidence 9 – Student Details

Show the student's profile image and details.

Evidence 10 – CRUD Verification

Show that Edit and Delete still work correctly after the change.

13. Implementation Summary

This change introduces a student profile image upload facility to the Admin Student Management module. Administrators can select and upload profile images in PNG, JPG, or JPEG format while creating a student record. The system validates the selected file,

associates the uploaded image with the correct student, and displays the image in the student management interface.

The change is designed to improve student identification and usability without changing the existing authentication and authorization model. Existing student CRUD operations remain available.

14. Final Status

Implementation Status: To Be Implemented

Priority: Medium

Difficulty: Medium

Change Type: Frontend + Backend Enhancement

Database Impact: One profile image reference field may be required.

Security Impact: File type validation and administrator authorization must be maintained.

Outstanding Issues: None at the planning stage.